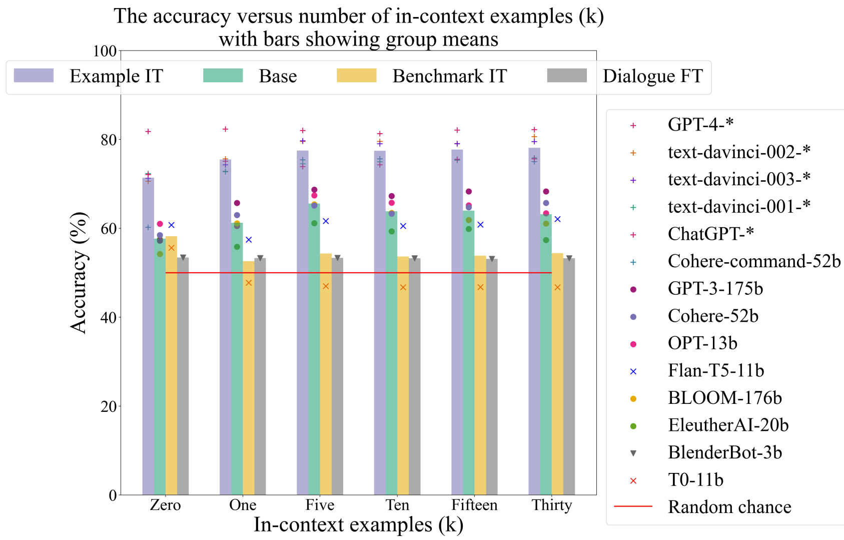

Figure 2: The few-shot accuracy for the best model of each class (e.g. the best performing model in the class Cohere-command is the 52b model, whereas the best model in the class OPT is the 13b model). The bars show the group means. Models fine-tuned on example-level instructions perform better than most other models, especially for $k > 0$. For all models there is a significant gap between best accuracy and human accuracy (which is 86.2%). * means size unknown.

Table 2: Scaling results for OpenAI's text-<engine>-001-series, for which we do not know the number of non-embedding parameters but do know the ordering in terms of size. The colors indicate whether going up in size (from left-to-right) increases performance significantly or not.

Engine	Ada	Babbage	Curie	Davinci
0-shot	56.5% $\pm$ 5.8	64.5% $\pm$ 1.8 (+8.0%)	69.0% $\pm$ 2.9 (+4.5%)	72.3% $\pm$ 2.8 (+3.3%)
5-shot	57.6% $\pm$ 2.8	66.1% $\pm$ 0.3 (+8.5%)	71.3% $\pm$ 1.3 (+5.2%)	74.5% $\pm$ 1.0 (+4.0%)

only looking at the best prompt type for each model class (i.e. structured or natural), the results remain that models in the group *Example IT* perform best.

Insight 3: Models instruction-tuned at the example-level have the most favourable scaling properties, but some base models also show positive correlation with scale. Figure 3 shows the scaling behaviour of the model classes for which we know the number of non-embedding parameters. We highlight 0- and 5-shot results, because for $k > 5$ the accuracy of most models plateaus (see Figure 2). However, detailed results for other k can be found in Appendix K.10. Note that we do not know the number of parameters for OpenAI's 'text-<engine>-001'-series, but we do know the order of the engines in size, and we separately present its scaling results in Table 2. Except OpenAI's 'text-<engine>-001'-series, none of the models show significant performance increase with model size for the 0-shot evaluations. However, for k-shot evaluations with $k \geq 1$ we observe significant positive correlation with size for the models in the *Example IT* class for which we have multiple sizes (Cohere-command and 'text-<engine>-001') as well as some models in the base model class. Not only do the models in the *Example IT* class exhibit higher performance for the same model size, these models also have a steeper performance increase with size than the base models. Comparing the scaling properties of the best base model (GPT-3) with Cohere-command, we see that the increase in performance from the second-largest to the largest model is 0.04% per billion parameters from GPT-3-6.7B to GPT-3-175B and 0.15% per billion parameters for Cohere-command-6B to Cohere-command-52b (exact numbers used to calculate the slope can be found in Appendix K.10). If performance is linearly extrapolated from this curve GPT-3 reaches human-level performance at 642b parameters where Cohere-command would need 125b parameters.

Insight 4: GPT-4 reaches average human-level performance with chain-of-thought prompting. For the model groups that benefit from in-context examples, we attempt to push performance further

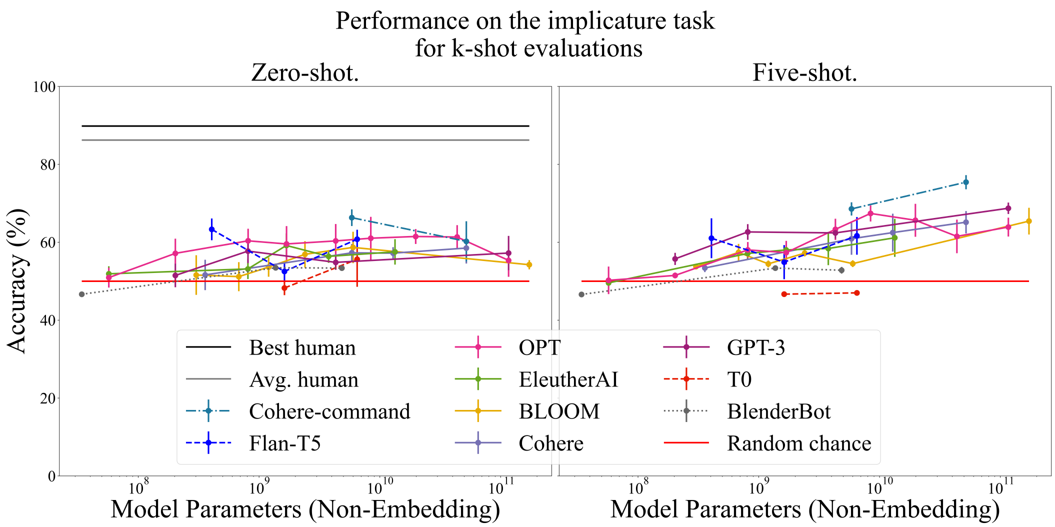

Figure 3: Scaling results for the model classes of which we know the number of non-embedding parameters. The error bars show standard deviation over prompt templates. Cohere's command models instruction-tuned at the example-level perform better than all other models. For all models there is still a significant gap between best accuracy and human accuracy.

Table 3: Results of the chain-of-thought (CoT) experiment for models in the group *Example IT*. The numbers between brackets show the difference in performance with the number on the same row one column to the left. Most models benefit from CoT-prompting, but not all. Additionally, GPT-4 reaches average human-level performance with CoT prompting. ⋆ means size unknown.

Model	**0-shot**	**5-shot**	**5-shot CoT**
text-davinci-001-⋆	72.3% ± 2.8	74.5% ± 1.0 (+2.2%)	67.3% ± 2.6 (-7.2%)
text-davinci-002-⋆	70.6% ± 2.3	79.6% ± 2.0 (+9.0%)	80.1% ± 0.8 (+0.5%)
text-davinci-003-⋆	71.2% ± 2.8	79.7% ± 0.6 (+8.5%)	83.6% ± 0.6 (+4.0%)
ChatGPT-⋆	72.1% ± 6.0	73.9% ± 6.3 (+1.8%)	77.2% ± 1.0 (+3.3%)
GPT-4-⋆	81.8% ± 1.8	82.0% ± 1.7 (+0.2%)	**86.5%** ± **1.0** (+4.5%)
Cohere-command-52b	60.2% ± 5.2	75.4% ± 1.8 (+15.2%)	75.3% ± 0.5 (-0.1%)

with chain-of-thought prompting. We manually write a five-shot chain-of-thought prompt for all six prompt templates, and evaluate model performance using this prompt. One of the six chain-of-thought prompts can be found in Table 4 in Appendix F, and the other five are provided in the supplementary material. We only present the results for the group Example IT here, since CoT prompting did not improve performance for two of the base model classes we tried (see Appendix K.7). Consequently, we decided not to apply this experiment to the other models in the base group to save compute costs. The results of are shown in Table 3. We find that chain-of-thought prompting does not help for all models, but is nonetheless able to boost performance of GPT-4 to 86.5% ± 1.0. This is on-par with average human-level performance, and slightly below human best performance at 89.8%. To illustrate how explicit reasoning helps implicature understanding, we highlight a CoT generated by GPT-4 for an example from the dataset that models persistently get wrong. "A: Is there a bus I can get to the station? B: You can't rely on it". The implicature is yes, there is a bus, you just cannot rely on it. GPT-4 five-shot gets this wrong for all six templates. With CoT it gets it right for five of six templates. The generated CoT for one template is the following:

> Alice says 'You can't rely on it.' Alice must be implying that there is a bus, but it may not be dependable or timely. This means the response to Bob's question is yes, but with a caution about reliability. Answer: yes

More completions can be found in Appendix J.

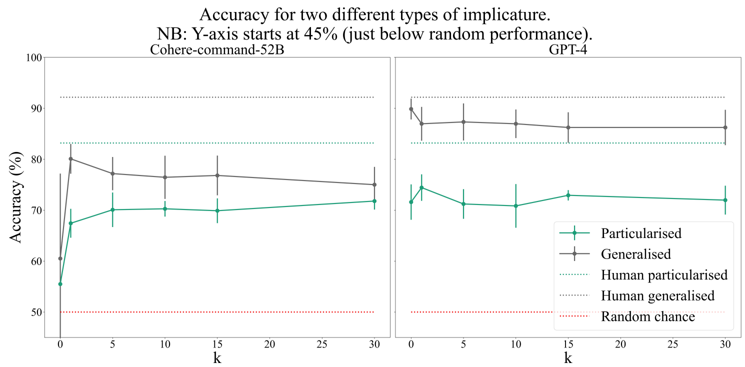

Figure 4: The accuracy v. k for the generalised and particularised examples obtained by the Example IT models Cohere-command and GPT-4. Particularised (context-heavy) examples are often significantly more difficult than generalised (context-free) examples for both models and humans. For most models, in-context prompting can mitigate this, but for others (like GPT-4), a significant gap remains. We see that Cohere-command-52b achieves similar performance as GPT-4 on the particularised examples, but significantly lower on the generalised examples.

Table 4: An example from the dataset for two types of implicature found in the test set. The rightmost column shows the amount of that type we manually found in the test set.

Type	**Example Utterance**	**Example Response**	**Impl.**	**#**
Generalised	You know all these people?	Some.	No.	47
Particularised	Want to stay for a nightcap?	I've gotta get up early.	No.	94

Insight 5: Models often struggle with the same type of examples humans struggle with. We manually labeled 217 examples of the 600 examples in the test set according to a taxonomy. The remaining 383 examples do not fall as clearly within a category and are grouped together as type *other*. In Table 4 the two types of examples that occur frequently in the dataset are exemplified. *Generalised* implicatures require little or no context to be understood. They are the simplest type of example in the test set, and generally imply the same thing ("some" almost always implies "not all"). *Particularised* implicatures, by contrast, do require context to be resolved. For example, from Table 4, we need the context that it is undesirable to stay up late drinking when one has to get up early (see in Appendix E for more on generalised vs. particularised). In these type of examples, the context needed to resolve it is different every time. We label three other types of implicatures in the dataset, but since the analysis of these examples does not show significant patterns, we present it in Appendix K.9. We show the accuracy broken down per example type for two models from the Example IT group, as these patterns hold more broadly for almost all models evaluated (see the detailed results broken down per example type in Appendix K.9). Figure 4 shows that for lower k, the models often have a significantly worse performance for particularised examples than for generalised examples, just like humans do. For some, like Cohere-command-52b, this is mitigated by few-shot prompting, which brings particularised and generalised performance closer together (sometimes at the cost of generalised performance). For others, like GPT-4, the gap between particularised and generalised performance remains large for all k. From the bottom row in Figure 4 we observe that the edge GPT-4 has over Cohere-command-52b seems mostly driven by a higher accuracy on generalised examples. The accuracy on the particularised examples is comparable between those two models.

5 Discussion

In this study we use prompting to evaluate whether different groups of LLMs can resolve implicatures. In designing our experimental protocol, we carefully considered various alternatives, and here we discuss limitations of the chosen approach. Firstly, evaluating LLM competencies is inherently uncertain and sensitive to prompt choice. Nonetheless, we are confident our evaluation is comprehensive enough to assess implicature understanding: we apply six different prompt templates per test example, each used in three different prompting techniques (zero-shot, few-shot, chain-of-thought). Addition-